

## **Drainage Report**

**Sumner-Bonney Lake  
School District  
New Roundabout  
199th Ave. Ct. E. at BLHS/MVMS**

**Prepared for:**

**Sumner-Bonney Lake School District  
19701 – 104<sup>th</sup> Street East  
Bonney Lake, WA 98391**

**Contact: Marina Tanay  
Phone: (253) 891-6305**

**Prepared by:  
Sitts & Hill Engineers, Inc.  
4815 Center Street  
Tacoma, Washington 98409**

**Contact: David Fillmore, P.E.  
Phone: (253) 474-9449**

**January 2026  
Job Number 20,931**



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## Project Engineers Certification

I hereby state that this Drainage Report for the **New Roundabout: 199<sup>th</sup> Ave. Ct. E. at BLHS/MVMS** has been prepared by me or under my supervision and meets the standard of care and expertise which is usual and customary in this community for professional engineers. I understand that neither Pierce County nor the City of Bonney Lake will assume liability for the sufficiency, suitability, or performance of drainage facilities prepared by me.



01/26/2026

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# 1 Proposed Project Description

The proposed public road improvements addressed in this report are associated with the new roundabout (RAB) at the intersection of 199<sup>th</sup> Ave. Ct. E. and the entrances to Bonney Lake High School (BLHS)/Mountain View Middle School (MVMS). The new RAB will include: two circulatory lanes, curb, gutter, and sidewalk along 199<sup>th</sup> Ave Ct E and on to BLHS/MVMS site. Additional ROW dedication is proposed along both streets to meet City and County requirements. Utility improvements are limited new traffic signals and illumination as well as some relocations of hydrants, as necessary to accommodate the proposed RAB. The disturbance to the area of offsite and on-site work will be covered by a single NPDES permit from Washington State Department of Ecology.

This report has been prepared in accordance with the 2021 Pierce County Stormwater Management and Site Development Manual (hereafter, 'Manual').

The project site is located along 199<sup>th</sup> Avenue Court East in Bonney Lake, between 107<sup>th</sup> Street Court East and MVMS southern driveway entrance. The MVMS address is 10921 199<sup>th</sup> Avenue Court East, Bonney Lake, Washington 98391. Please see the Vicinity map included as Figure 1 for the public improvements area. The adjacent property of the proposed RAB is currently developed as a school site ("Public Facilities" zoning per Bonney Lake) with one existing driveway to BLHS and two existing driveway to MVMS. All other adjacent developments are single family residences ("Suburban Residential" per Pierce County).

The existing high point in the 199<sup>th</sup> Ave Ct E (located north of the BLHS/MVMS intersection) separates the stormwater runoff in the public Right-of-Way (ROW) into two separate drainage basins; north drainage basin drains to an infiltration trench (located on 104<sup>th</sup> St E) and south drainage basin drains to an infiltration pond (located on the southeastern portion of 198<sup>th</sup> Ave Ct E/112<sup>th</sup> St E). The project also includes area disturbance on the BLHS and MVMS site. Stormwater runoff will continue to discharge into the onsite stormwater management facilities. The total disturbed area project site is 110,400 square-feet (2.53 acres). The disturbance includes 82,000 square-feet (1.88 acres) of new plus replaced hard surfaces. See Exhibit 1 in Appendix A depicting coverages in the tables below.

*Table 1.0 – Surface Coverage Table*

| Drainage Basin  | PGHS   |      | NPGHS  |      | Pervious |      | Total   |      |
|-----------------|--------|------|--------|------|----------|------|---------|------|
|                 | S.F.   | Ac.  | S.F.   | Ac.  | S.F.     | Ac.  | S.F.    | Ac.  |
| <b>Existing</b> | 55,400 | 1.27 | 8,900  | 0.20 | 46,100   | 1.06 | 110,400 | 2.53 |
| North           | 12,200 | 0.28 | 1,800  | 0.04 | 9,500    | 0.22 | 23,500  | 0.54 |
| South           | 21,700 | 0.50 | 3,700  | 0.08 | 20,400   | 0.47 | 45,800  | 1.05 |
| BLHS            | 11,600 | 0.27 | 2,500  | 0.06 | 14,200   | 0.33 | 28,300  | 0.65 |
| MVMS            | 9,900  | 0.23 | 900    | 0.02 | 2,000    | 0.05 | 12,800  | 0.29 |
| <b>Proposed</b> | 70,400 | 1.62 | 11,600 | 0.27 | 28,400   | 0.65 | 110,400 | 2.53 |
| North           | 12,500 | 0.29 | 2,200  | 0.05 | 1,900    | 0.04 | 16,600  | 0.38 |
| South           | 45,700 | 1.05 | 7,200  | 0.17 | 19,400   | 0.45 | 72,300  | 1.66 |
| BLHS            | 8,800  | 0.20 | 1,900  | 0.04 | 6,400    | 0.15 | 17,100  | 0.39 |
| MVMS            | 3,400  | 0.08 | 300    | 0.01 | 700      | 0.02 | 4,400   | 0.10 |

*PGHS = Pollution Generating Hard Surface*

*NPGHS = Non-Pollution Generating Hard Surface*

Table 1.1 – Surface Coverage Table (cont.)

| Drainage Basin | New+Replaced HS |      | Basin Net Change |       |
|----------------|-----------------|------|------------------|-------|
|                | S.F.            | Ac.  | S.F.             | Ac.   |
| <b>Total</b>   | 82,000          | 1.88 | 0                | 0.00  |
| North          | 14,700          | 0.34 | -6,900           | -0.16 |
| South          | 52,900          | 1.21 | 26,500           | 0.61  |
| BLHS           | 10,700          | 0.25 | -11,200          | -0.26 |
| MVMS           | 3,700           | 0.08 | -8,400           | -0.19 |

HS = Hard Surface

Figure 1 - Vicinity Map



Near Map, 2025

199<sup>th</sup> Ave Ct E, Bonney Lake, WA 98391



## **1.1 Minimum Requirements**

Please see the Pierce County Flow Charts, included as Figure 2 in this report, for determining the requirements for this project. The offsite improvements addressed in this report are part of the larger private development. Because the total new hard surface area is more than 5,000 square feet, Minimum Requirements #1-10 apply to all new and replaced hard surfaces and converted vegetation areas. Below is a summary of how the project will satisfy each of the Minimum Requirements.

### **1.1.1 Minimum Requirement #1 – Preparation of Stormwater Site Plans**

This report and the accompanying drawings satisfy this requirement.

### **1.1.2 Minimum Requirement #2 - Construction Stormwater Pollution Prevention**

Please see the Construction Stormwater Pollution Prevention Plan (CSWPPP), and Demolition and TESC Plans, Notes, and Details. The CSWPPP is prepared in accordance with Section 2.2 in Volume II of the Manual and includes a discussion of the 13 required elements and reference to any applicable BMPs are included in the CSWPPP.

### **1.1.3 Minimum Requirement #3 - Source Control of Pollution**

Source control BMPs for construction sites are included in the Construction Stormwater Pollution Prevention Plan (CSWPPP), Maintenance and Source Control Manual, and the project plans. These documents are bound under separate covers and included in this permit submittal.

### **1.1.4 Minimum Requirement #4 - Preservation of Natural Drainage Systems and Outfalls**

As mentioned in the Section 1 above, the existing high point in the 199<sup>th</sup> Ave Ct E (located north of the BLHS/MVMS entrances) separates the stormwater runoff in the public ROW into two separate drainage basins as well as disturbance in the BLHS and MVMS drainage basin. The stormwater runoff from the north drainage basin is tributary to an existing infiltration trench located at the south along 104<sup>th</sup> St E. The runoff from the north drainage basin is tributary to the existing infiltration pond located southeast of 198<sup>th</sup> Ave Ct E/112<sup>th</sup> St E intersection. The runoff from the BLHS and MVMS drainage basin is tributary to their respective onsite stormwater management facilities. Please see Appendix A, Exhibit 3 for Downstream Map.

In the proposed condition, the stormwater management facilities serving the north, BLHS, and MVMS drainage basin will continue fully infiltrate the replaced hard surfaces without any negative impact (no significant increase in stormwater runoff). The County infiltration pond serving south drainage basin will fully infiltrate the stormwater runoff generated from the new and replaced hard surfaces. The natural drainage patterns in the vicinity of the project site will be maintained and stormwater discharge from the proposed improvements will not be altered from the natural locations.

### **1.1.5 Minimum Requirement #5 - On-site Stormwater Management**

The project triggers Minimum Requirements #1-10 and is not located outside the UGA on a parcel larger than 5 acres. Therefore, meeting LID Performance Standard through the use of any Flow control BMPs and Soil Preservation and Amendment Best Management Practice (BMP) are required to comply with this minimum requirement.

The LID Performance Standard will be met by full infiltration from the existing infiltration trench (north drainage basin) and infiltration pond (south drainage basin). It is proposed to follow the LID Performance Standard and provide Soil Preservation and Amendment BMP on

all pervious site surfaces. Please refer to the Appendix A, Exhibit 3 for the locations of the existing stormwater management facilities, and project plans for the specific locations of these Best Management Practices (BMPs).

#### **1.1.6 Minimum Requirement #6 – Runoff Treatment**

Since the stormwater runoff from the ROW improvements is generated in two separate basins that are hydraulically independent, Minimum Requirement #6 is considered in each threshold Discharge Area (TDA) separately. Although the pollution generating hard surfaces proposed in each TDA exceed the threshold, the project area tributary the north drainage basin will be replaced hard surface area and will not result in a significantly increase in stormwater runoff. The project area tributary to the south drainage basin includes new and replaced hard surface area; however, the existing infiltration pond was designed to accommodate for the full build-out of 198<sup>th</sup> Ave Ct E/199<sup>th</sup> Ave Ct E under 'CRP 5676' and has adequate capacity to provide runoff treatment for the proposed improvements. As a result, no additional runoff treatment is proposed, and will continue to discharge to the existing stormwater facilities. Refer to Section 9 of this report for additional discussion of water quality.

#### **1.1.7 Minimum Requirement #7 – Flow Control**

There is more than 10,000 square-feet of effective hard surfaces associated with the project; therefore, flow control is required per Section 2.4.7 of Volume 1 of the Manual. In the existing condition, stormwater runoff generated by the project site discharges to an infiltration trench (north drainage basin) and an infiltration pond (south drainage basin). Stormwater runoff is fully infiltrated in these areas. In the proposed condition, stormwater runoff from the new and replaced hard surfaces will continue to fully infiltrate through the existing stormwater management facilities without any negative impact in the proposed condition. The infiltration pond was sized for 100% infiltration for previously approved road improvements and future road improvements of 199<sup>th</sup>/198<sup>th</sup> Ave Ct E and 112<sup>th</sup> St E. Please see previously approved "19<sup>th</sup> Avenue East – CRP 5676" Drainage Report (prepared by Goldsmith Land Development Services, dated December 2012), provided under separate cover. See Appendix A for Basin Map, demonstrating the proposed surface area coverage is within the design capacity of the infiltration pond.

#### **1.1.8 Minimum Requirement #8 – Wetlands Protection**

Stormwater runoff generated by the project site does not directly or indirectly discharge into a wetland; therefore, Wetland Protection is not required for this project.

#### **1.1.9 Minimum Requirement #9 – Operation and Maintenance**

Please see the Maintenance and Source Control Manual submitted as a separate document. This manual was prepared using the Pierce County's recordable template and in accordance with Section 3.3.6 in Volume I of the Manual and includes the maintenance checklists for the stormwater management facilities proposed in the public ROW.

#### **1.1.10 Minimum Requirement #10 – Financial Liability**

The project owner is a public entity, Sumner-Bonney Lake School District; therefore, Pierce County does not require performance bonding.

Figure 2 - Minimum Requirements Flow Charts

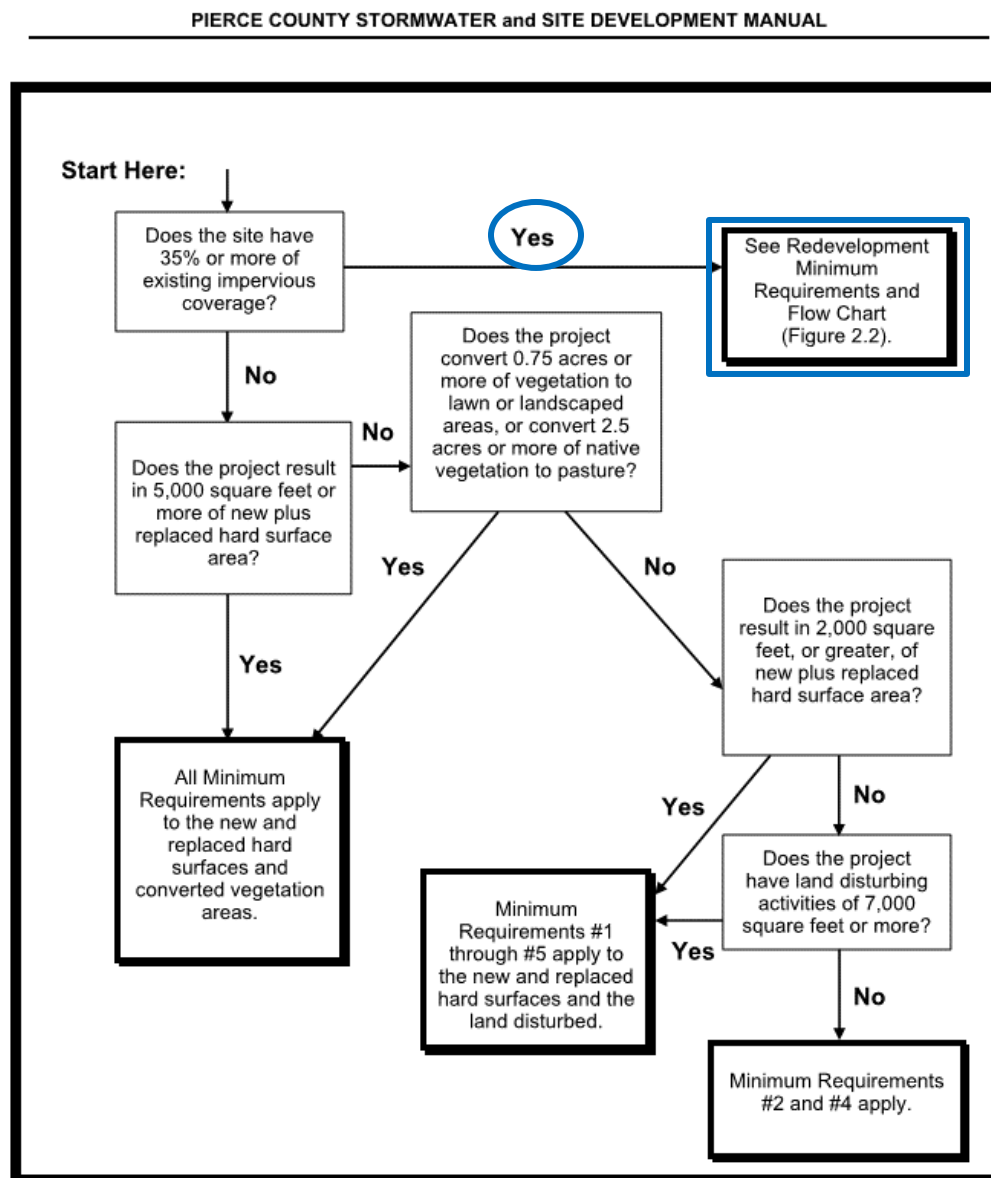


Figure 2.1. Flow Chart for Determining Requirements for New Development.

PIERCE COUNTY STORMWATER and SITE DEVELOPMENT MANUAL

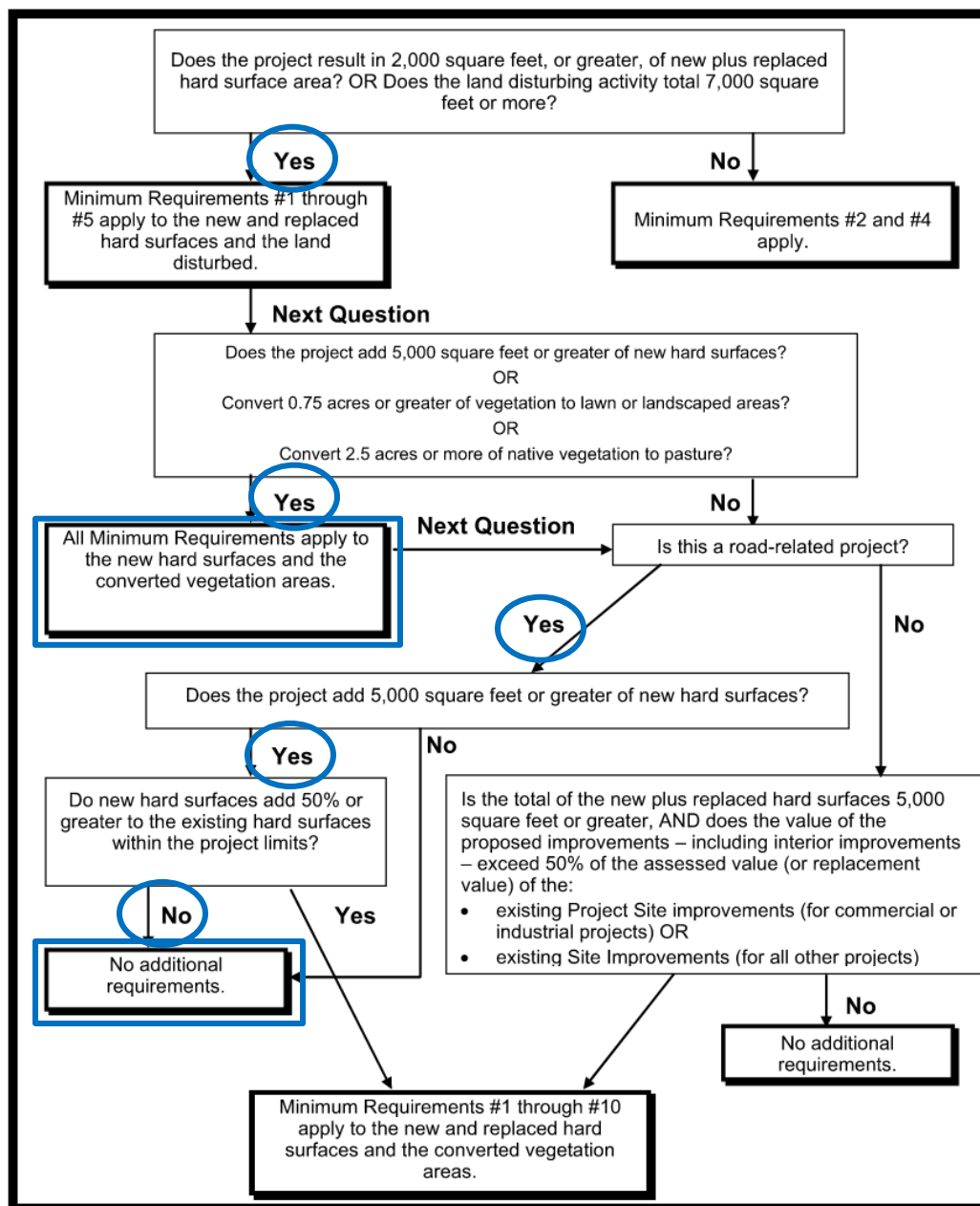


Figure 2.2. Flow Chart for Determining Requirements for Redevelopment.

Figure 3 - Minimum Requirement #5 Flow Chart

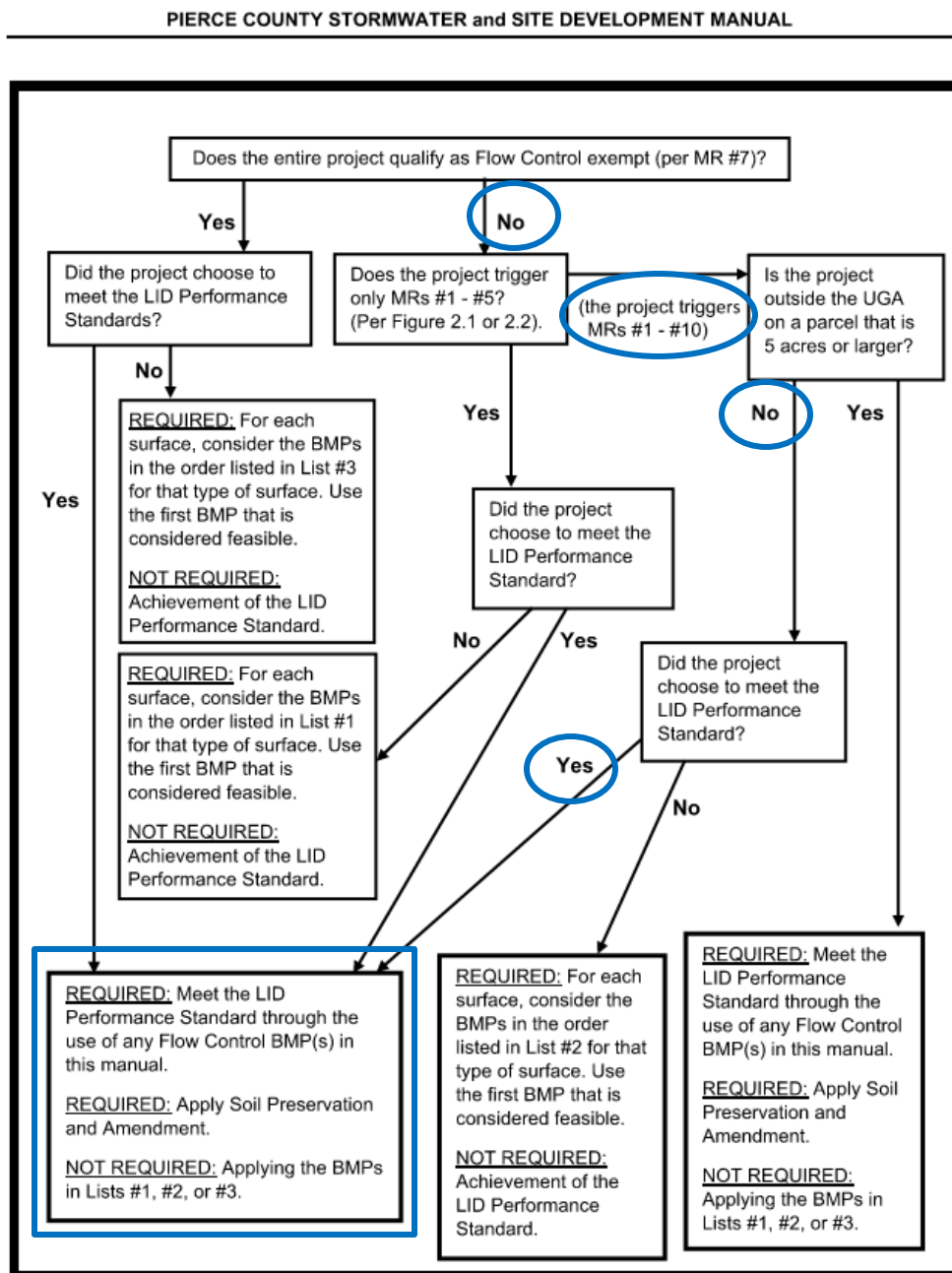


Figure 2.3. Flow Chart for Determining Minimum Requirement #5 Requirements

## **2 Existing Conditions**

The project limits consist of approximately 110,400 square-feet (2.53 acres) of disturbed area.

Between 107<sup>th</sup> Street Court East and the southern driveway entrance of Mountain View Middle School, 199<sup>th</sup> Avenue Court East is currently developed as three-lane roadway with concrete sidewalk shoulder and an intersection providing access to Bonney Lake High school (to the west) and Mountain View Middle School (to the east). The high point of 199<sup>th</sup> Avenue Court East exist at the north of the intersection. Currently, stormwater runoff from the northern half of the roadway is collected by a catch basin located at southeast intersection of 107<sup>th</sup> Street Court East and is ultimately discharging to an existing infiltration trench. Stormwater runoff from the southern half sheet flows south along the curb and gutter and is collected by catch basins and discharges to an existing infiltration pond. Stormwater runoff is fully infiltrated through both stormwater management facilities, and no drainage or flooding issues are known to exist.

According to Pierce County property information, the site lies within, or contains, an Aquifer Recharge Area. According to the FEMA FIRMette included in Figure 4 the site is situated in Zone X, an area of minimal flood hazard. We are unaware of any environmentally hazardous areas or historical drainage problems on the project site. There are no proposed or existing slopes steeper than 2H:1V (ditches) in the project area.

## **3 Infiltration Rates / Soils Report**

A Geotechnical Engineering Report has been submitted for this project under a separate cover.

## **4 Wells and Septic Systems**

According to DOE Well Report, there are no existing nor proposed wells and septic systems located within ROW improvements limits. We are not aware of septic systems located in the adjacent properties.

## **5 Fuel Tanks**

We are not aware of fuel tanks located above or below grade on the project site.

## **6 Sub-basin Description**

The project site is divided into four drainage basins based on existing topography; two within public ROW (north and south) and two in the school sites (BLHS and MVMS). In existing conditions, stormwater runoff from the south drainage basin is conveyed via sheet flow and catch basins toward the existing County infiltration pond. Stormwater runoff from the north basin drains independently from the south basin which will be directed to the infiltration trench. Stormwater runoff from BLHS/MVMS frontage are collected on the onsite catch basins and discharged to their respective onsite stormwater management facilities.

Under proposed conditions, stormwater runoff from four drainage basins will continue to discharge to their respective stormwater management facilities. The north, BLHS, and MVMS drainage basin will have a net decrease in new plus replaced hard surface. The BLHS drainage basin will be evaluated under a separate stormwater report, as onsite improvements are proposed. However, the south drainage basin will have an increase of 26,500 square-feet (0.61 acres) of new plus replaced hard surface. The existing County infiltration pond has been evaluated to accommodate the increase of hard surfaces and will continue to fully treat and

infiltrate the proposed improvements. Please see Section 9 for the evaluation of the existing County infiltration pond.

The four drainage basins are hydraulically independent and no runoff conveyance between basins are anticipated. The existing stormwater management facility of each drainage basin will continue to function as the primary facilities for their respective drainage areas.

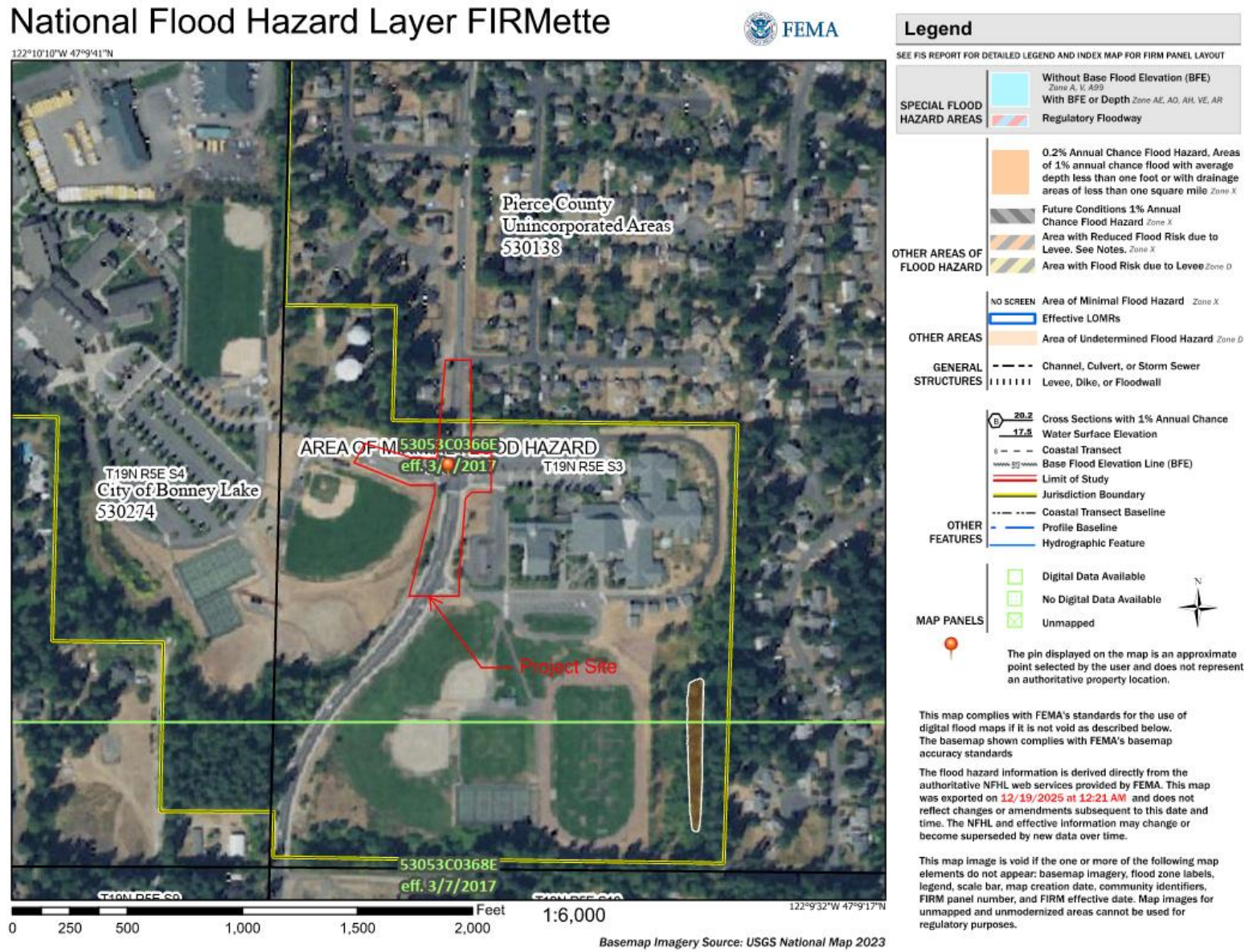
## **7 Flood Plain Analysis**

According to the FEMA FIRMette, included as Figure 4 in this report, the site is located within Zone X, an area of minimal flood hazard.

## **8 Aesthetic Considerations for Facilities**

Not applicable – stormwater management facilities are not proposed in this project.

Figure 4 - FEMA Flood Map



## **9 Facility Sizing and Downstream Analysis**

Proposed stormwater facilities include a conveyance system, comprised of underground piping and catch basins. The proposed conveyance system can adequately convey the 100-year, 24-hour event per the conveyance analysis provided in Appendix B.

The stormwater infiltration pond (south basin) was designed by Gold Smith Land Development Services (see Appendix C for a snippet of the previously approved “198<sup>th</sup> Avenue East – CRP 5676” report) and were sized for 100% infiltration and water quality treatment using the Western Washington Hydrology Model (WWHM), a DOE-approved continuous runoff model. In the design, the pond is sized to accommodate a full build-out of 199<sup>th</sup> Ave Ct E that is comprised four to five travel-lanes. Currently, the existing corridor is consisting of two-lane road (198<sup>th</sup> Ave Ct E/199<sup>th</sup> Ave Ct E) from Rhodes Lake Rd E north to 104<sup>th</sup> St E, with a turn lane to BLHS/MVMS site. Water quality treatment is through a natural soil layer in the infiltration pond bottom with a suitable CEC.

Based the previously approved report, the Central Drainage extends from Sta. 35+20 south to approximately Sta. 65+40 (high-point), for a total distance of about 3,020 feet. The proposed RAB project’s “south drainage basin” is wholly within this basin. With the addition of two lanes (11 feet width), the existing infiltration pond is capable of accommodating an additional 66,440 square-feet (1.53 acres) of hard area. The South Drainage Basin proposes an additional 26,500 square-feet (0.61 acres) of hard area to discharge into the infiltration pond; therefore, no stormwater management facilities or water quality treatment are proposed.

## **10 Utilities**

Proposed stormwater utilities will be installed so as not to interfere with the existing storm, natural gas, water, sanitary sewer and electric utilities. Please see the Grading and Drainage Plans for the location of the existing and proposed utilities and profiles for the utility crossing information.

## **11 Covenants, Dedications, Easements**

After construction, operation and maintenance of the stormwater management system will be the responsibility of the owner, City of Bonney Lake.

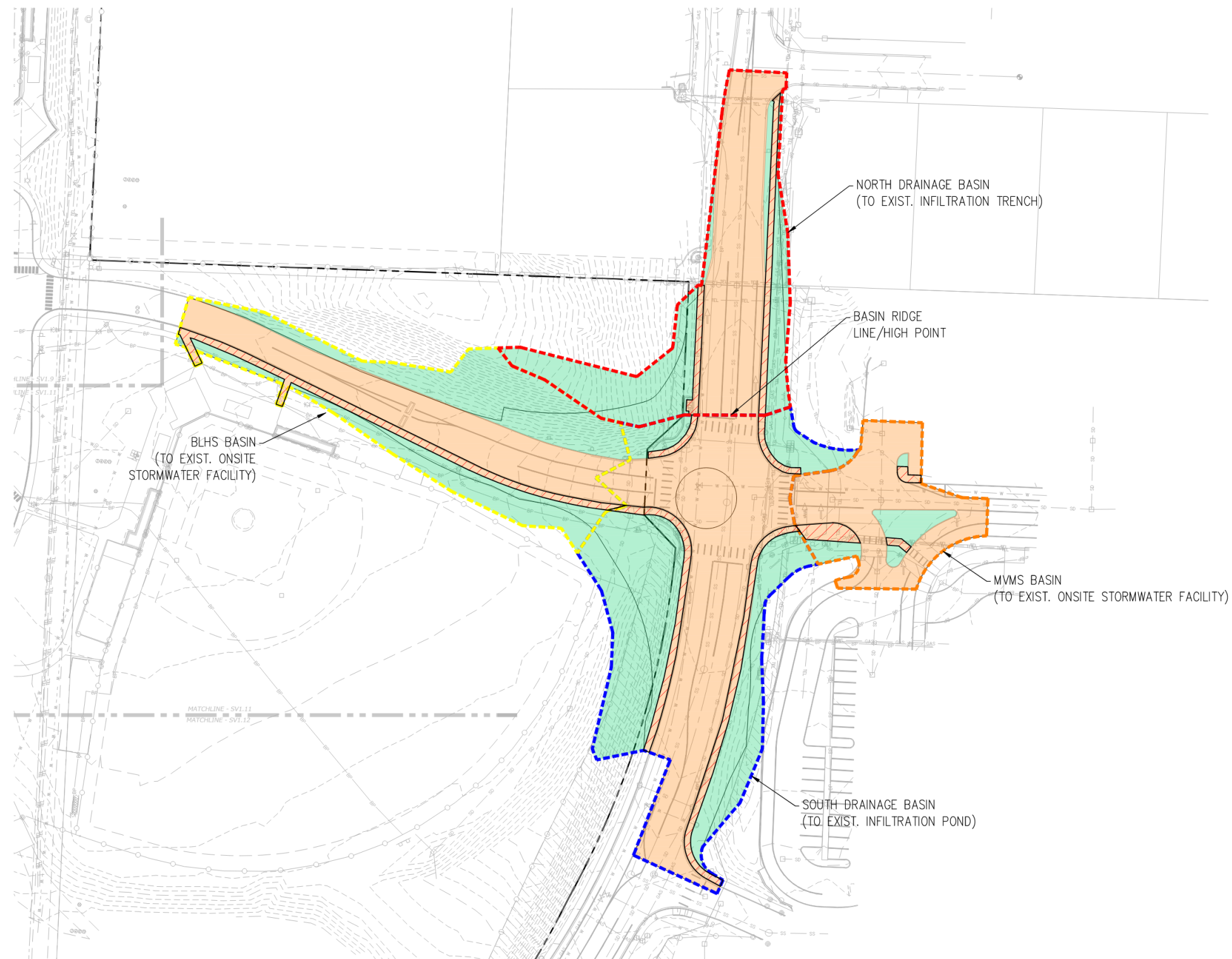
## **12 Property Owner’s Association Articles of Incorporation**

There are no property owner’s association articles of incorporation.

## **13 Other Permits or Conditions Placed on the Project**

There are no other conditions placed on this project.

## **Appendix A      Subbasin Maps**

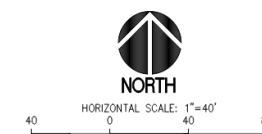


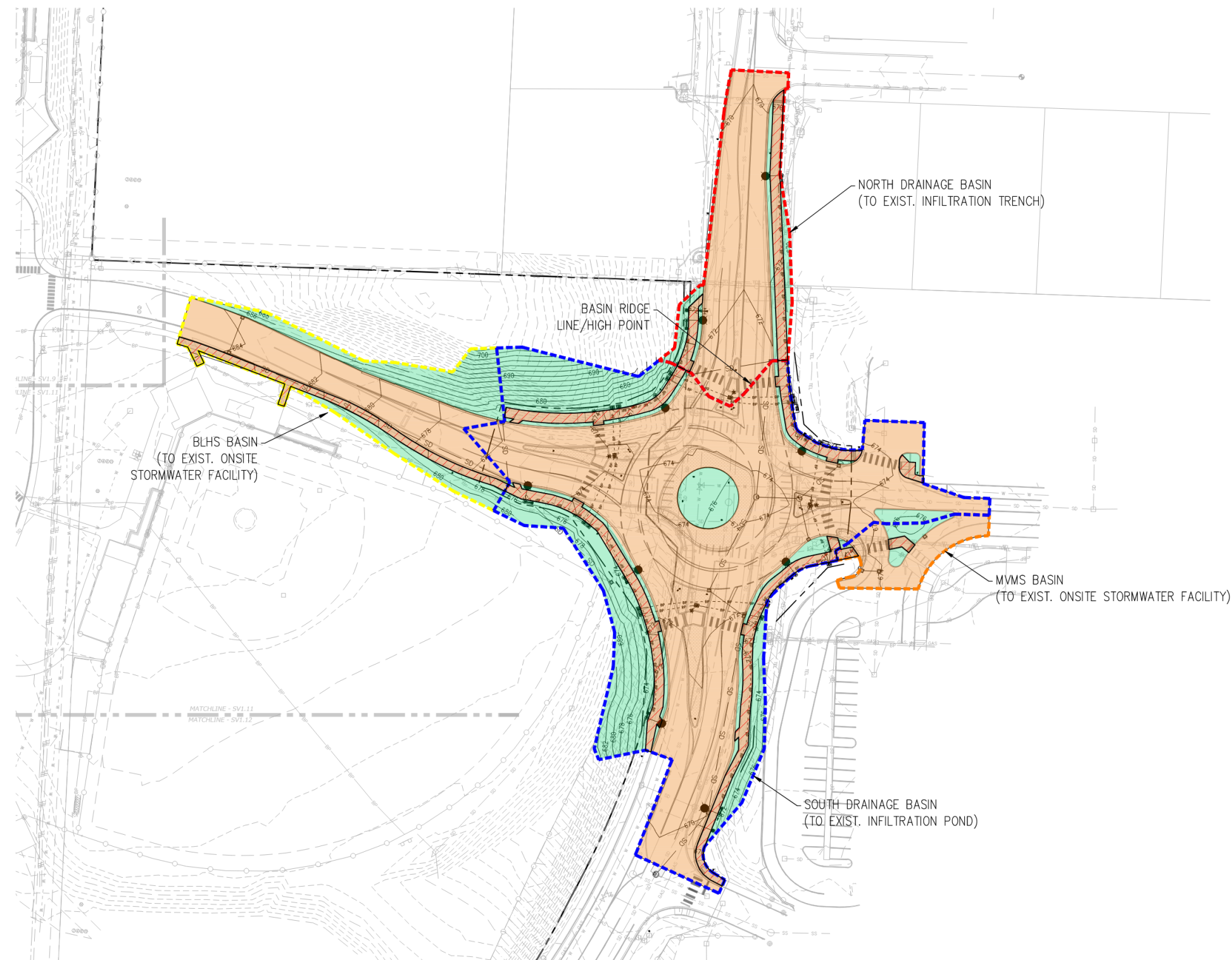
## LEGEND

|                                                                                     |                            |
|-------------------------------------------------------------------------------------|----------------------------|
| <u>PROJECT LIMITS: 110,400 SF (2.53 AC)</u>                                         |                            |
|  | NORTH: 23,500 SF (0.54 AC) |
|  | SOUTH: 45,800 SF (1.05 AC) |
|  | BLHS: 28,300 SF (0.65 AC)  |
|  | MVMS: 12,800 SF (0.29 AC)  |
| <u>PGHS: 55,400 SF (1.27 AC)</u>                                                    |                            |
|  | NORTH: 12,200 SF (0.28 AC) |
|                                                                                     | SOUTH: 21,700 SF (0.50 AC) |
|                                                                                     | BLHS: 11,600 SF (0.27 AC)  |
|                                                                                     | MVMS: 9,900 SF (0.23 AC)   |
| <u>NPGHS: 8,900 SF (0.20 AC)</u>                                                    |                            |
|  | NORTH: 1,800 SF (0.04 AC)  |
|                                                                                     | SOUTH: 3,700 SF (0.08 AC)  |
|                                                                                     | BLHS: 2,500 SF (0.06 AC)   |
|                                                                                     | MVMS: 900 SF (0.02 AC)     |
| <u>PERVIOUS: 46,100 SF (1.06 AC)</u>                                                |                            |
|  | NORTH: 9,500 SF (0.22 AC)  |
|                                                                                     | SOUTH: 20,400 SF (0.47 AC) |
|                                                                                     | BLHS: 14,200 SF (0.33 AC)  |
|                                                                                     | MVMS: 2,000 SF (0.05 AC)   |

## EXHIBIT 1 - PROJECT SITE EXISTING SURFACE COVERAGE

SCALE: 1"=40'





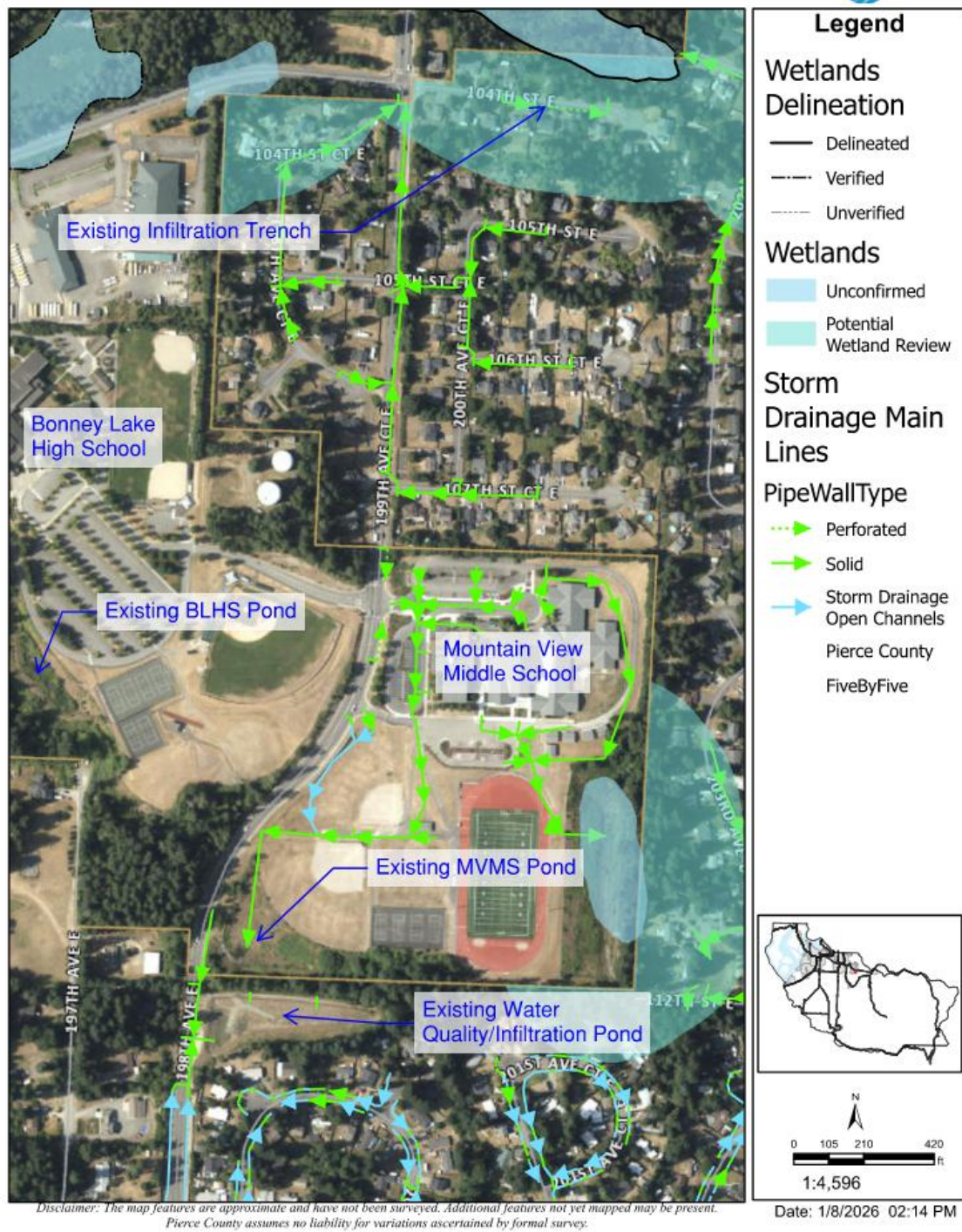
## LEGEND

|  |                                      |
|--|--------------------------------------|
|  | PROJECT LIMITS: 110,400 SF (2.53 AC) |
|  | NORTH: 16,600 SF (0.38 AC)           |
|  | SOUTH: 72,300 SF (1.66 AC)           |
|  | BLHS: 17,100 SF (0.39 AC)            |
|  | MVMS: 4,400 SF (0.10 AC)             |
|  | PGHS: 70,400 SF (1.62 AC)            |
|  | NORTH: 12,500 SF (0.29 AC)           |
|  | SOUTH: 45,700 SF (1.05 AC)           |
|  | BLHS: 8,800 SF (0.20 AC)             |
|  | MVMS: 3,400 SF (0.08 AC)             |
|  | NPGHS: 11,600 SF (0.27 AC)           |
|  | NORTH: 2,200 SF (0.05 AC)            |
|  | SOUTH: 7,200 SF (0.17 AC)            |
|  | BLHS: 1,900 SF (0.04 AC)             |
|  | MVMS: 300 SF (0.01 AC)               |
|  | PERVIOUS: 28,400 SF (0.65 AC)        |
|  | NORTH: 1,900 SF (0.04 AC)            |
|  | SOUTH: 19,400 SF (0.45 AC)           |
|  | BLHS: 6,400 SF (0.15 AC)             |
|  | MVMS: 700 SF (0.02 AC)               |

EXHIBIT 2 - PROJECT SITE PROPOSED SURFACE COVERAGE  
SCALE: 1"=40'



## PublicGIS



### EXHIBIT 3 - DOWNSTREAM MAP

## Appendix B      Engineering Calculations

### Conveyance Calculations Summary

12-inch diameter storm drain piping is the minimum pipe size diameter required by Pierce County development standards. The minimum proposed slope is 0.50%. To demonstrate that 12-inch pipe at 0.50% will adequately handle tributary runoff, the entire project area were assumed tributary to a 12" pipe at 0.50%. Per the calculation below (for the larger basin), the proposed storm drains are adequately sized.

| SBUH Hydrograph Synthesis |           |         |           |         |                 |
|---------------------------|-----------|---------|-----------|---------|-----------------|
| <i>Input:</i>             |           |         |           |         |                 |
| Pervious Area             | 0.650     | (acres) |           | CN      | 24-hour<br>86.0 |
| Impervious Area           | 1.880     | (acres) |           | CN      | 98.0            |
| Total Area:               | 2.53      | (acres) |           | S       | 1.63            |
| Time of Concentration     | 6.0       | (min.)  |           | S       | 0.20            |
|                           |           |         |           | w       | 0.4545          |
|                           | 6-month   | 2-year  | 10-year   | 25-year | 100-year        |
| Precipitation (in.)       | 1.28      | 2.0     | 3.0       | 3.5     | 4.1             |
| <i>Results:</i>           |           |         |           |         |                 |
| Event                     | Peak Flow | Volume  | Peak Time |         |                 |
| 6-month, 24-hour          | 0.59      | 8,093   | 7.67      |         |                 |
| 2-year, 24-hour           | 1.03      | 14,112  | 7.67      |         |                 |
| 10-year, 24-hour          | 1.68      | 22,814  | 7.67      |         |                 |
| 25-year, 24-hour          | 2.00      | 27,243  | 7.67      |         |                 |
| 100-year, 24-hour         | 2.40      | 32,598  | 7.67      |         |                 |

| Mannings Equation- Circular Pipe                                                                                                                                      |          |       |                    |                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|--------------------|--------------------------------------------------------------------|
| <b>Instructions:</b> Input pipe information in cells highlighted gray.<br>Unhighlighted cells are intermediate computations. Final solution is highlighted in yellow. |          |       |                    |                                                                    |
| Pipe Diameter                                                                                                                                                         | D        | 12.0  | (inches)           |                                                                    |
| Manning's "n"                                                                                                                                                         | n        | 0.013 |                    |                                                                    |
| Slope                                                                                                                                                                 | S        | 0.50  | (%)                | <b>Formulas:</b>                                                   |
| Flow Depth                                                                                                                                                            | d        | 10.80 | (inches)           |                                                                    |
| Flow Depth Ratio                                                                                                                                                      | d/D      | 0.90  | (%)                | $d/D = (d/D)$                                                      |
| Flow Point Angle                                                                                                                                                      | $\theta$ | 286.3 | (deg)              | $\theta = 2 * \cos^{-1}((1 - (2 * d/D)))$                          |
| Flow Area                                                                                                                                                             | A        | 0.745 | (ft <sup>2</sup> ) | $A = 1/8 * (\text{RADIANS}(\theta_r) - \sin(\theta_r)) * (D/12)^2$ |
| Wetted Perimeter                                                                                                                                                      | WP       | 2.498 | (ft)               | $WP = \pi * (D/12) * \theta_r / (2 * \pi)$                         |
| Hydraulic Radius                                                                                                                                                      | R        | 0.298 | (ft)               | $R = A / WP$                                                       |
| Flow                                                                                                                                                                  | Q        | 2.692 | (cfs)              | $Q = (1.49/n) * A * R^{2/3} * S^{0.5}$                             |
| Velocity                                                                                                                                                              | V        | 3.616 | (fps)              | $V = Q / A$                                                        |

12-inch diameter storm drain pipe sloped at 0.5% is able to convey 2.692 cfs flowing at 90% depth which is larger than the anticipated 100-year flow of 2.40 cfs tributary to the south drainage basin; therefore, the proposed and reconfigured conveyance system is adequately sized.

## Appendix C      '198<sup>th</sup> Avenue East CRP 5676' Drainage Report

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# **198<sup>th</sup> Avenue East CRP 5676**

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## Drainage Report

*Prepared for: Pierce County for support of CRP 5676*

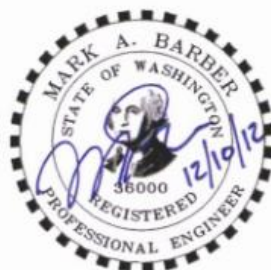
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Job Number: 11110



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I hereby state that this Drainage Report for the 198<sup>th</sup> Ave. East – CRP 5676 has been prepared by me or under my supervision and meets the standard of care and expertise which is usual and customary in this community for professional engineers. I understand that Pierce County does not and will not assume liability for the sufficiency, suitability, or performance of drainage facilities prepared by me.



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## Appendices

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*see 5674 for general comments*

## SECTION 1: PROJECT DESCRIPTION

The Cascadia Employment Based Planned Development Planned Unit Development, now known as Tehaleh, was approved by Pierce County Hearing Examiner on June 18, 1999. The conditions of approval require constructed improvements along 198th Avenue East from 199th Avenue Court (in the City of Bonney Lake) south to the Cascadia Boulevard roundabout for mitigation of Phase 1 of the EBPC traffic impacts. The off-site road project is divided into the following three separate County Road Projects:

- County Road Project 5676: 104th Street East south to Rhodes Lake Road East.
- County Road Project 5674: Rhodes Lake Road East south to 120th Street East.
- County Road Project 5675: 120th Street East to 144th Street East (to Cascadia Boulevard Roundabout).

A majority of the existing corridor is a two (2) lane road that will ultimately be improved to four (4) or five (5) lanes throughout to accommodate the full build-out of Tehaleh (formerly Cascadia). To serve Tehaleh's first phase, however, a lesser scope of interim improvements is required as described below.

### Pierce County Road Project 5676

This project is Phase 1 interim improvements for Pierce County Road Project (CRP) 5676 as required mitigation for the Cascadia EBPC. CRP 5676 lies within the jurisdiction of Pierce County and the City of Bonney Lake and proposes to construct/reconstruct a north/south two (2) lane road (199th Avenue Court East / 198th Avenue East) from 104th Street East in the City of Bonney Lake south to Rhodes Lake Road East, a distance of approximately 3,350 feet. The ultimate roadway configuration includes retaining walls. The interim alignment and road section, however, has been designed such that no walls are required. The intent is to defer the retaining walls until the ultimate widening is necessary.

This roadway segment is designed for two (2) lanes with additional turn lane channelization at the Bonney Lake High School / Mt. View Junior High entrance and Rhodes Lake Road E. The portion in the City of Bonney Lake has: curb, gutter, and sidewalk; both sides. The portion in Pierce County has: barrier curb, gutter, and paved/ gravel shoulder. In addition, the project includes a stormwater facility, clearing, grading, and relocation of existing utilities.

The proposed stormwater control plan for CRP 5676 will utilize a closed conveyance system that directs stormwater runoff to a proposed water quality/ infiltration facility located within a closed depression at the south east corner of the intersection of 198<sup>th</sup> East and 112<sup>th</sup> St. E. Utilizing the closed depression will reduce the extent of required grading for the facility. Water quality treatment is proposed through a natural soil layer in the infiltration pond bottom with a suitable CEC.



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## **SECTION 2: EXISTING CONDITIONS**

The existing 198<sup>th</sup> Ave E. roadway within the area included in CRP 5676 consists of a 2 lane paved road with a gravel edge. Under existing conditions approximately 3,350 ft of 198<sup>th</sup> Ave E. drain towards a closed depression at the southeast intersection of 198<sup>th</sup> Ave E and 112<sup>th</sup> St. E. Under existing conditions drainage is conveyed mainly via ditches; however, there are areas that are poorly graded without a ditch section.

## **SECTION 3: INFILTRATION RATES/SOILS REPORT**

Geotechnical Report – 198th Avenue East Roadway Improvements – CRP 5676 104th Street East to Rhodes Lake Road East Pierce County, Washington Project No. T-5632-18, Terra Associates, Inc. August 1, 2006, updated April 30, 2012) – Included in Appendix E.

## **SECTION 4: WELLS AND SEPTIC SYSTEMS**

NA - This section is not applicable to the current application for construction of a Pierce County road project.

## **SECTION 5: FUEL TANKS**

NA - This section is not applicable to the current application for construction of a Pierce County road project.

## **SECTION 6: SUB-BASIN DESCRIPTION**

Drainage sub-basin areas are shown on the enclosed full size Figures 2, 6 and 7. These figures show the overall drainage area for the proposed stormwater control facility and the individual conveyance system inlet areas analyzed in Section 9 of this report.

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## SECTION 7: ANALYSIS OF THE 100-YEAR FLOOD

NA - This section is not applicable to the current application for construction of a Pierce County road project.

## SECTION 8: AESTHETIC CONSIDERATION FOR FACILITIES

The proposed road and the proposed stormwater facility have been designed and located to minimize the impacts of the project on adjacent properties including aesthetic considerations. The proposed infiltration facility has been designed to conform to the available space for the facility and has been designed to minimize the extent of proposed grading to minimize the visual impact of the facility.

## SECTION 9: FACILITY SIZING AND DOWNSTREAM ANALYSIS

### SECTION 9.1 Drainage Control Plan

The 198th Ave. corridor, from 104th Ave. to 120th Ave. has been separated into 3 drainage basins. The North Drainage Basin is from 104th Ave. south to a high point at approximately Sta. 35+20. The Central Drainage Basin is from Sta. 35+20 south to a high point approximately Sta. 65+40 and the South Drainage Basin from Sta. 65+40 south to a pond to be constructed in CRP 5675. The stormwater flow control and treatment facility for the central drainage basin will be constructed with CRP 5676. The majority of the proposed roadway corridor within this basin is unimproved and therefore has no existing drainage facilities.

In the developed condition, stormwater runoff from the proposed roadway, including a portion 112<sup>th</sup> Ave. will drain to a low point at approximately Sta. 48+00. East of this low point is an existing open space tract, Tract A, of the plat of Rhododendron Park. This tract is at the natural low point of the surrounding area and a closed depression. This tract is located at the southeast corner of the intersection of 198<sup>th</sup> Ave E and 112<sup>th</sup> St. E. as shown in the enclosed full size Figure 2. This location has been selected as the location for the proposed infiltration and water quality treatment facility for CRP 5676. The facility has been designed to infiltrate 100% of the inflow volume. This facility is sized to accommodate both the currently proposed road improvements, but also the potential ultimate future road improvements to 198<sup>th</sup> Ave E. and 112<sup>th</sup> St. E. The facility has a pre-settlement cell and has been designed to treat runoff by utilizing the existing natural soil layer at the bottom of the facility that meets CEC requirements for treatment through the soil layer.

### Performance Standard

The proposed stormwater infiltration facility has been designed in accordance with the *Pierce County 2008 Stormwater Management and Site Development Manual (SMSDM)*.

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## Land Surface Cover

The drainage basin area draining to the proposed stormwater facility as shown in the enclosed Figure 2 has the following land surface area break down. These areas reflect not only the currently proposed improvements but also potential future road improvements. SCS soils mapping (Appendix D) identifies the soil type for the drainage area draining to the CRP 5676 facility as Everett soils, a SCS Hydrologic Group Type A soils. However, to be conservative in the sizing of the proposed infiltration facility SCS Hydrologic Group Type C soils have been selected for modeling of pervious area runoff. Type C soils generates greater runoff compared to Type A soils.

### Basin Land Surface Summary :

|                                                        |             |
|--------------------------------------------------------|-------------|
| Existing Impervious Area:                              | 3.33 acres  |
| Existing Pervious Areas (Grass)                        | 10.38 acres |
| Proposed New Impervious Area-198 <sup>th</sup> Ave. E. | 6.33 acres  |
| Proposed New Impervious Area-112 <sup>th</sup> St. E.: | 1.12 acres  |
| Proposed New Pervious Area (Grass)                     | 6.26 acres  |

## Infiltration Rate

The design infiltration rate is based on infiltration tests completed by the project geotechnical engineer. The project geotechnical report detailing the selection of the design infiltration rate has been included in Appendix E. For the proposed facility a design infiltration rate of 0.8in/ hr is recommended for the proposed infiltration facility.

## Methodology

Hydrologic Modeling was completed using the program WWHM 3, version 1.0 (06/30/2011). As specified in the 2008 Pierce County SMSDM, WWHM can be used for the analysis and design of stormwater flow control and treatment facilities provided the extended precipitation time series files developed for Pierce County are used. The East 44 WDM files were used for this project.

## Design Summary:

Detailed hydrologic modeling data and results are included in Appendix C. The following is a summary of the analysis results:

### Design Summary:

|                                         |              |
|-----------------------------------------|--------------|
| Design Infiltration Rate                | 0.8" / hr.   |
| Required Infiltration Storage Volume:   | 5.17 ac.-ft. |
| 100-year Water Surface Elev:            | 605.97 ft.   |
| Pond Crest Elev:                        | 610.00 ft.   |
| Available Freeboard:                    | 4.03 ft.     |
| 100-year Peak Inflow:                   | 6.90 cfs     |
| Required Water Quality Volume:          | 1.05 ac.-ft. |
| Required Pre-settlement Storage Volume: | 0.32 ac.-ft. |

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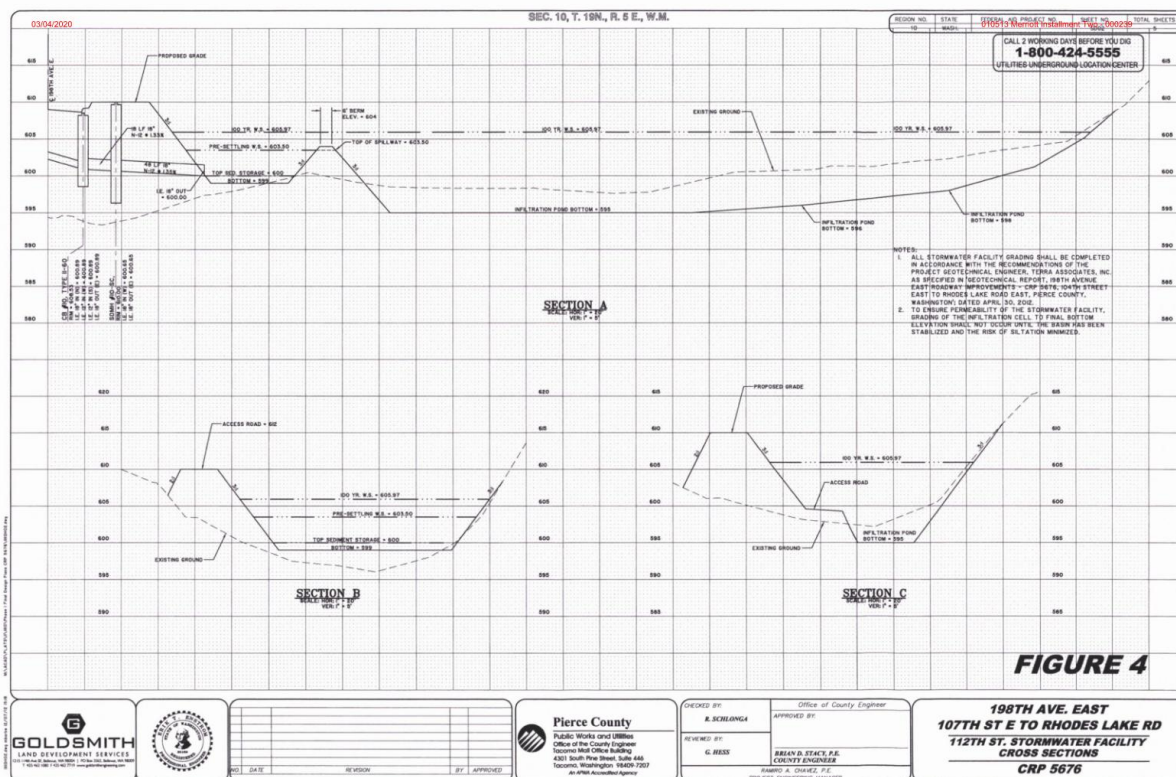
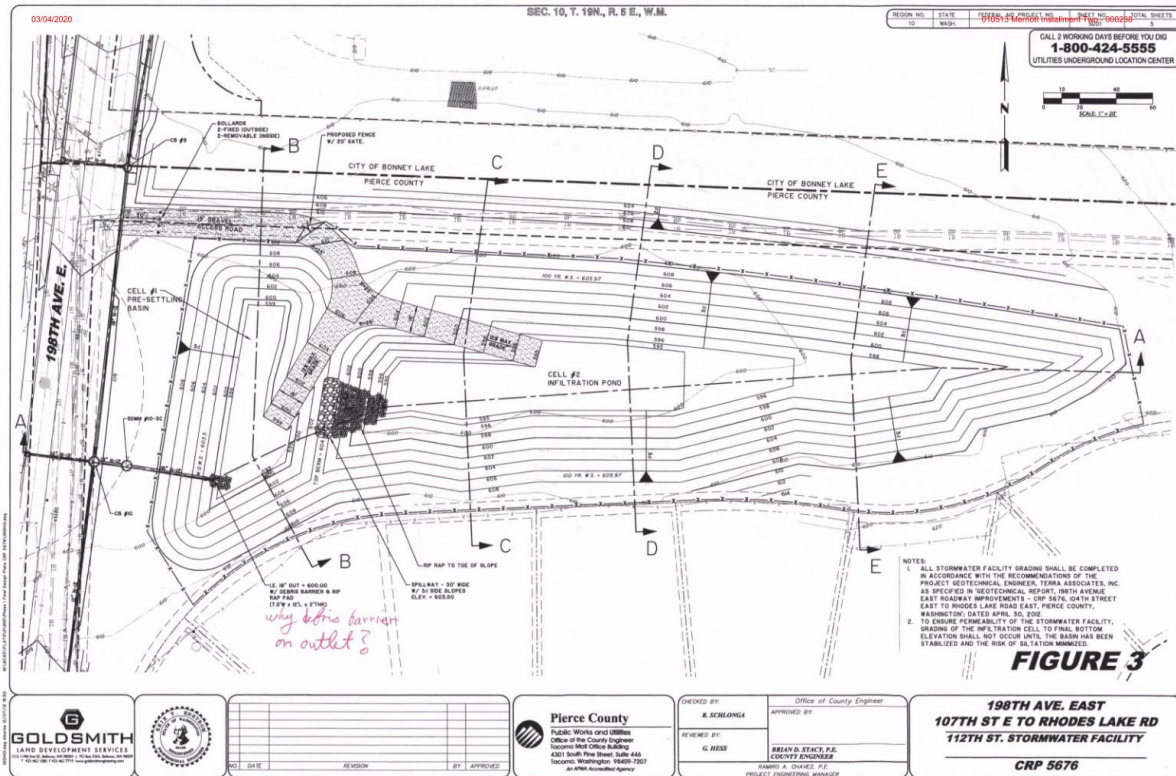
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**Pre-Settlement Pond: Stage – Area / Storage Table**

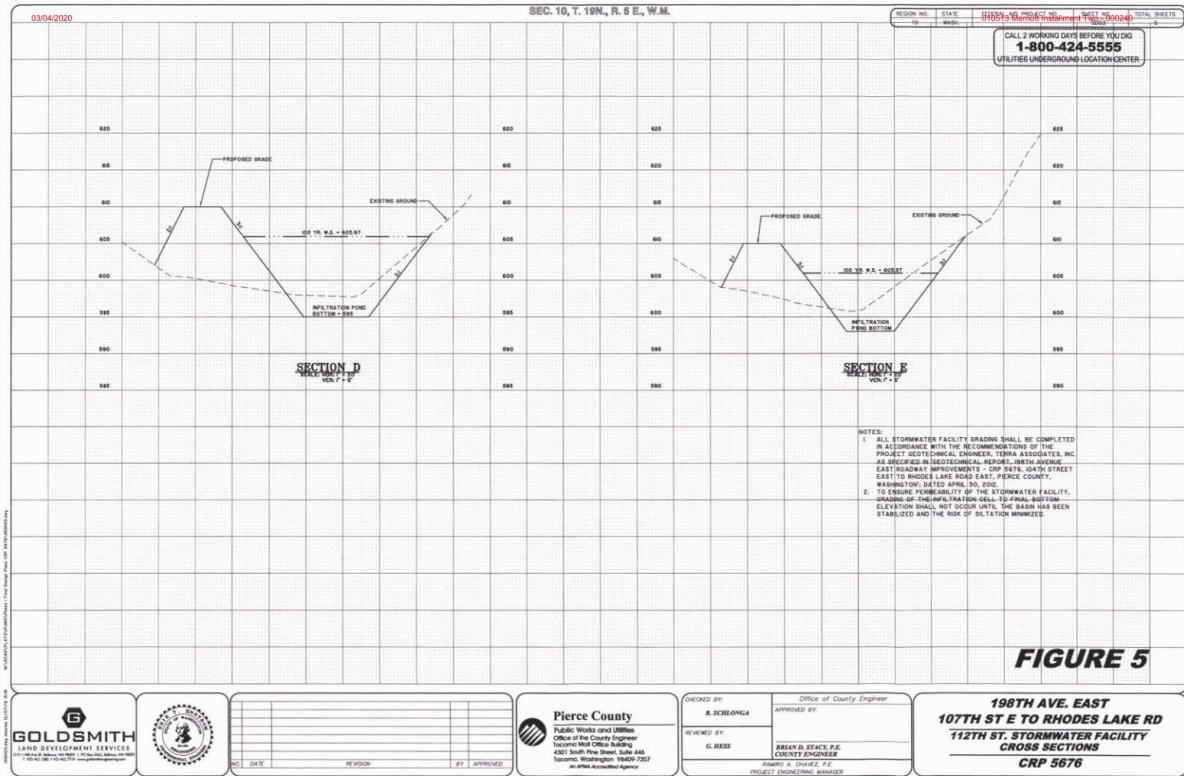
| Elevation (ft.)                                   | Area (sq. ft.) | Storage (cu. ft.) | Storage (ac. ft.) |
|---------------------------------------------------|----------------|-------------------|-------------------|
| 599                                               | 2,315          | --                | --                |
| 600                                               | 3,245          | 2,780             | 0.06              |
| 602                                               | 5,344          | 11,369            | 0.26              |
| 603                                               | 6,524          | 17,303            | 0.40              |
| 604                                               | 7,762          | 24,446            | 0.56              |
| Storage from elev. 600 to 603.5 ft. = 0.42 ac.ft. |                |                   |                   |

**Infiltration Pond: Stage – Area / Storage Table**

| Elevation (ft.)                                    | Area (sq. ft.) | Storage (cu. ft.) | Storage (ac. ft.) |
|----------------------------------------------------|----------------|-------------------|-------------------|
| 595                                                | 5,120          | --                | --                |
| 596                                                | 7,838          | 6,479             | 0.15              |
| 598                                                | 12,651         | 26,968            | 0.62              |
| 600                                                | 17,223         | 56,842            | 1.30              |
| 602                                                | 22,373         | 96,438            | 2.21              |
| 604                                                | 27,979         | 146,790           | 3.37              |
| 604.01                                             | 35,741         | 147,109           | 3.38              |
| 606                                                | 43,775         | 226,227           | 5.19              |
| 608                                                | 51,462         | 321,464           | 7.38              |
| Storage from elev. 595 to 605.97 ft. = 5.17 ac.ft. |                |                   |                   |



# SBLSD New Roundabout: 199th Ave. Ct. E. at BLHS/MVMS – Drainage Report



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## SECTION 9.2 Conveyance System Design Analyses

1. The proposed stormwater conveyance system for 198<sup>th</sup> Ave. E. has been designed to not only for the currently proposed road section, but also to accommodate the ultimate future roadway configuration and improvements. The design and hydraulic analysis has been completed in accordance with the *Pierce County 2008 Stormwater Management and Site Development Manual (SMSDM)*. The results of the conveyance analysis contained herein demonstrate that the proposed design meets these standards. Profiles showing the hydraulic grade line are contained herein. Detailed model data and output are included in Appendix C of this report.

The design analysis included the following assumptions:

2. The 2008 SMSDM requires that all conveyances elements within public roadways, or rights-of-way are required to convey the 25-year, 24-hour storm event when utilizing a uniform flow analysis. However, the 2008 SMSDM also specifies that Pierce County may require analysis of the conveyance system under backwater conditions using both the 25-year and 100-year, 24-hour storm events. The design proposed for 198<sup>th</sup> Ave E is based on a backwater analysis of the proposed conveyance system. The 2008 SMSDM requires a minimum freeboard of 6-inches under the 25-year event and limited overtopping of the system during the 100-year event as described in Volume III Section 4.4 of the 2008 SMSDM.
3. Peak flow rates from contributing drainage areas are based on the fully developed condition. The fully developed condition is taken as the ultimate future construction of full improvements of 198th Ave. East. For this analysis sub-basin areas have been uniformly modeled as 75% impervious and 25% grassed areas based on an analysis of land surface cover for the entire basin.
4. Drainage areas and the storm sewer layout are shown on the engineering plans and on the Storm Conveyance System Plan. (see enclosed full size Figures 3 and 4).
5. Peak design flow values are based on the Rational Method and the following criteria:
  - Runoff coefficients per SCS TR-55 methodology using a C=0.25 (lawn and grass) and C = 0.9 (impervious).
  - Time of Concentration conservatively based on minimum value of 5.0 minutes.
6. The peak rainfall intensity, IR, for the design storm event have determined using a unit peak rainfall intensity factor IR which is calculated from the appropriate Tc and coefficients based on the projects location. The coefficients used were from the IDF curves (Zone 1) as given in the 1998 Pierce County Stormwater Management and Site Development Manual Appendix E.
7. Mannings roughness coefficients; n, = 0.014 for concrete and CPEP smooth interior pipe.
8. The backwater analysis has been completed using Autodesk Storm and Sanitary Analysis 2012 (V 7.1.2186.1).
9. The tail water boundary condition is set by the pond peak water surface elevation of 604.40 ft for the 25-year event and 605.97 ft. for the 100-year event.

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## ***Conveyance System Analysis***

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### **Profiles**

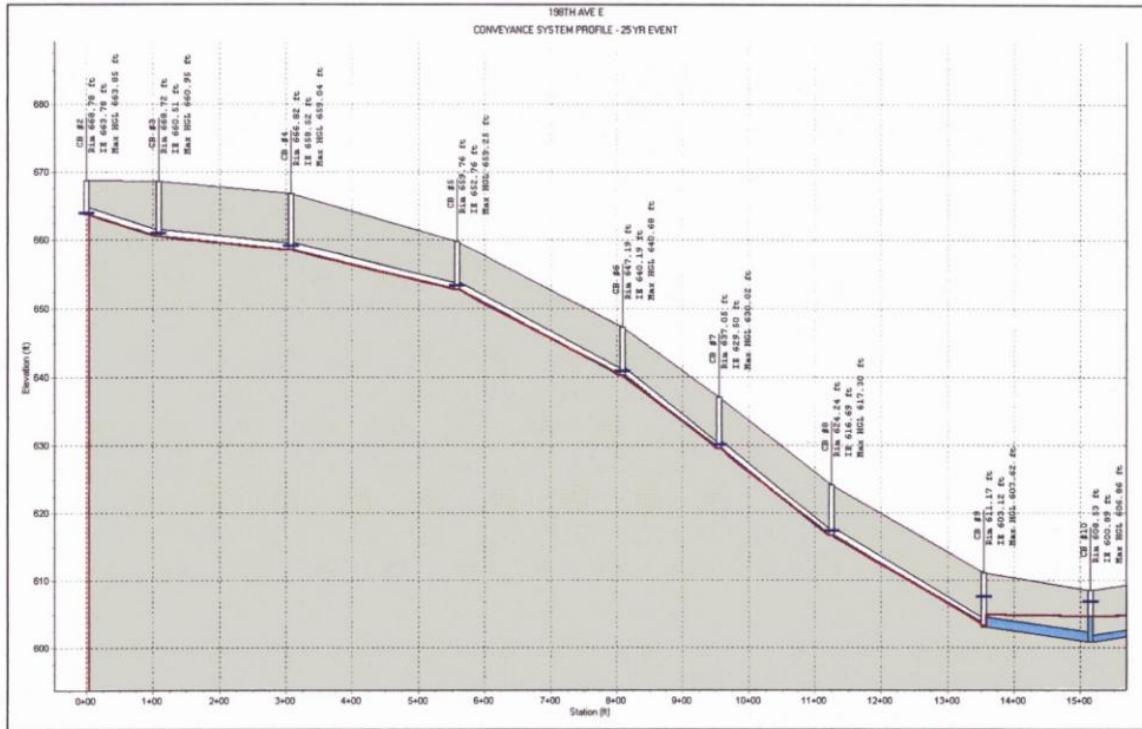
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LAND DEVELOPMENT SERVICES

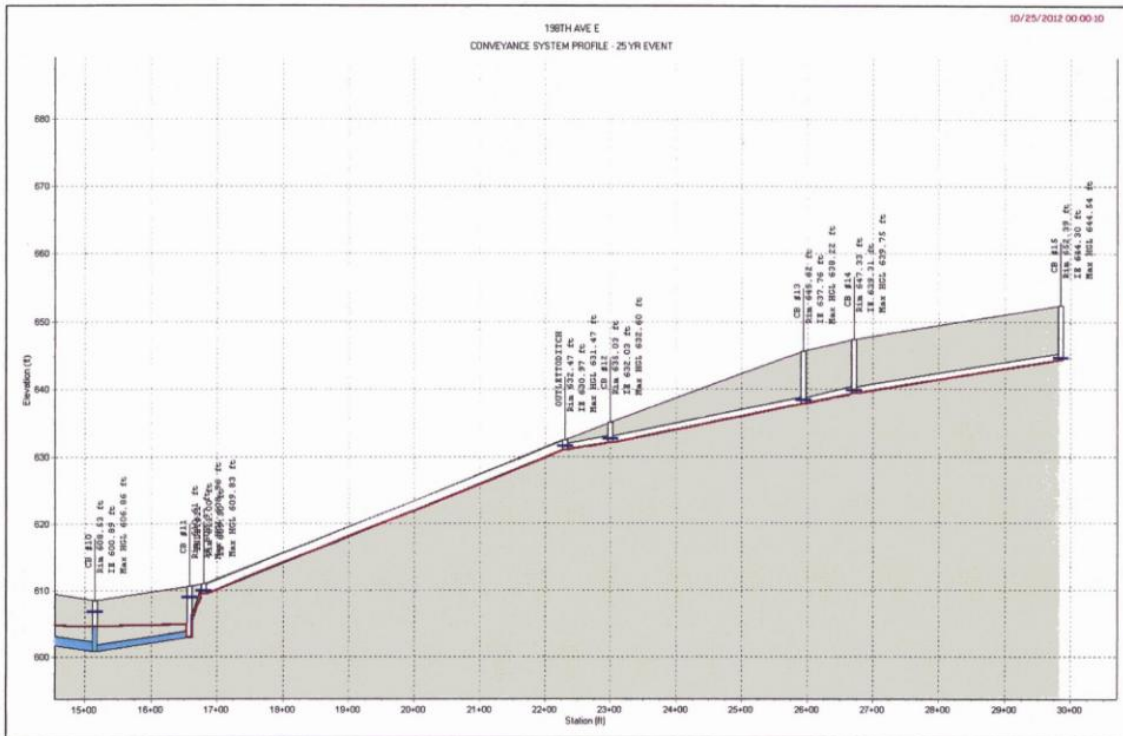
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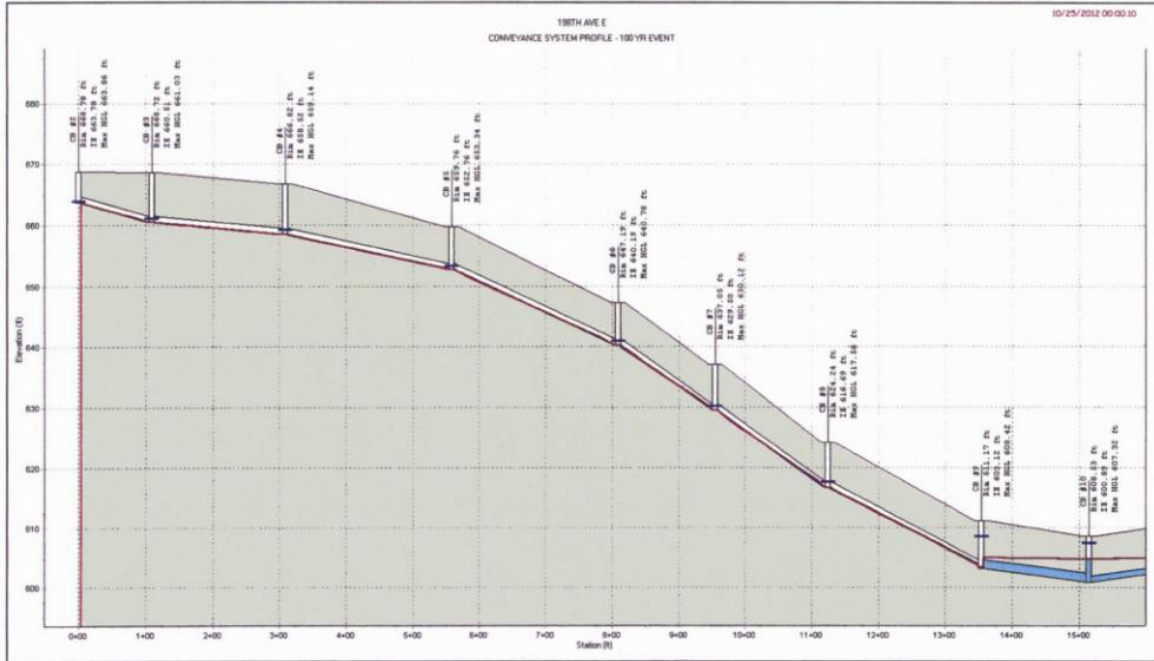
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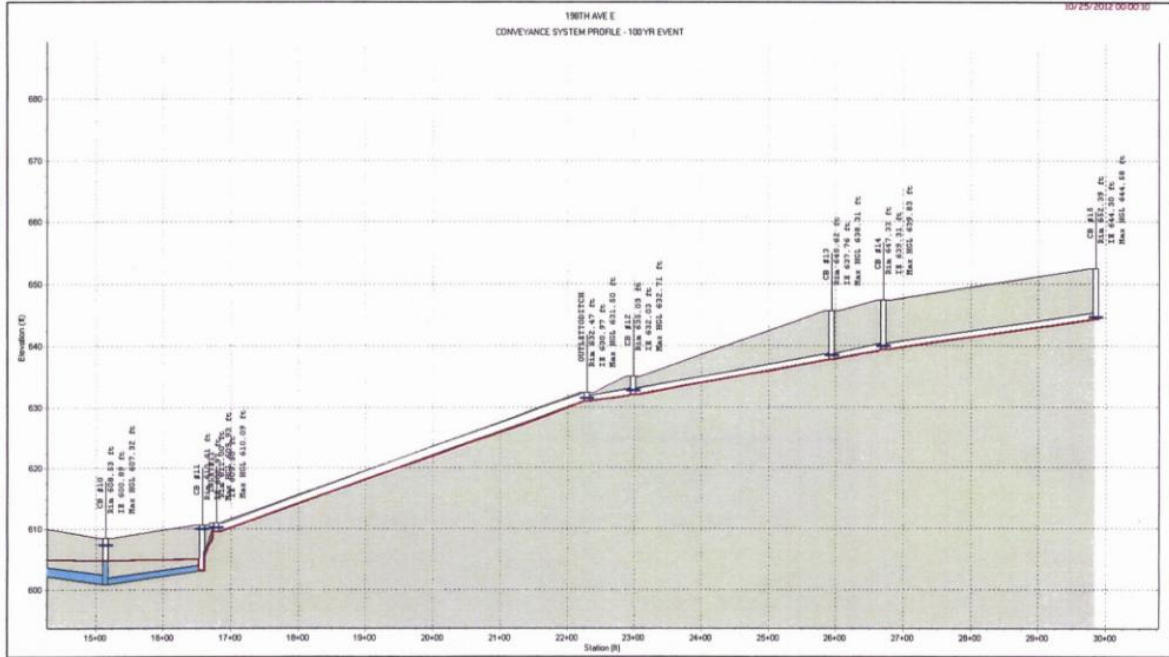
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**SECTION 10: COVENANTS, DEDICATIONS, & EASEMENTS**

NA - This section is not applicable to the current application for construction of a Pierce County road project.

**SECTION 11: PROPERTY OWNERS ASSOCIATION ARTICLES OF INCORPORATION**

NA - This section is not applicable to the current application for construction of a Pierce County road project.

**SECTION 12: OTHER PERMITS OR CONDITIONS PLACED ON THE PROJECT**

No additional permits are required and no additional conditions need to be met for this project.